Semester V **B. Tech CSE (AI)**

BCSAI 503: SIGNAL AND IMAGE PROCESSING

3L + 0T + 0P + 3C **MM 100**

Unit 1: Introduction

Fundamental steps in DIP, elements of DIP, Simple image model, sampling & quantization, basic relationships between pixels, colour image model.

Unit 2: Image Transforms

One-dimensional & two-dimensional DFT, cosine, sine, Hadamard, Haar, and Slant & KL transforms. Image Enhancement: Introduction, point operations, histogram modelling, spatial operations, Transform operations.

Unit 3: Image Restoration

Introduction, image observation models, Inverse & Wiener filtering, difference between enhancement & restoration Restoration-spatial filtering, Noise reduction in frequency domain.

Unit 4: Image Compression

Introduction, Pixel coding, Predictive coding, Transform coding, Inter-frame coding

Unit 5: Image Segmentation

Introduction, Spatial feature extraction, Transforms features, Edge detection, Boundary extraction, Segmentation techniques.

Text/Reference Books:

1. Digital Image Processing Using MATLAB, Gonzalez, Woods and Eddins, Gatesmark Publishing
2. Applications of Pattern Recognition, Fu, K.S., CRC Press
3. Digital Image Restoration, Andrews, H.C. Hunt, B.R., Prentice Hall, Englewood Cliffs.
4. Applications of Digital Signal Processing, Oppenheim, A.V.,Prentice Hall Englewood Cliffs.
5. Digital Image Processing, Gonzalez, R.C. & Wintz, P.A., Reading, Addison-Wesley.